

Matrix Theory: Lecture 15

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Stochastic Matrices

$x_j^{(t)}$ = probability of state j at time t

p_{ij} = probability of transitioning from state j to state i

$p_{ij} \geq 0$ (probabilities are non-negative)

$\sum_i p_{ij} = 1$ (probability of transitioning from state j to any state)

Probability of state j after transition:

$$x_i^{(t+1)} = p_{i1}x_1^{(t)} + p_{i2}x_2^{(t)} + \dots + p_{in}x_n^{(t)}.$$

In matrix form:

$$X^{(t+1)} = PX^{(t)}.$$

Is there always a stable distribution? $PX = X$? That is, is there a non-negative eigenvector with eigenvalue 1?



Postive Matrices (and Vectors)

Def. A is a **positive matrix** $A > O$ if $a_{ij} > 0$ for all i, j .
 $A > B$ if $A - B > O$.

Similarly A **non-negative matrix** $A \geq O$ if $a_{ij} \geq 0$ for all i, j .

Def. $|A|$ = matrix with elements $|a_{ij}|$.

Inequalites of numbers can be extended:

$$|X + Y| \leq |X| + |Y|, \quad |\alpha X| \leq |\alpha| |X|.$$



Postive Matrices

Theorem 10.14

Let $A > O$ be a positive matrix and $X \geq O$ be non-zero vector. Then

- ① $AX > O$;
- ② A cannot be nilpotent.
- ③ If $AX \geq \alpha X$ for some positive α then $|\lambda| \geq \alpha$ for at least one eigenvalue λ . The inequality is strict if $AX > \alpha X$.
- ④ If Y is any complex vector then $|AY| \leq A|Y|$. The equality is valid iff there exists a non-negative vector P and a real α such $Y = e^{i\alpha}P$.



Positive Matrices

Lemma

Suppose $A > 0$ with eigenvalues $\lambda_1, \dots, \lambda_n$.

If $\rho = \max_i |\lambda_i|$ and $AY = \lambda_k Y$ with $|\lambda_k| = \rho$ then

$$A|Y| = \rho|Y| \quad \text{and} \quad |Y| > 0.$$



Perron Theorem

Theorem 10.14 (Perron)

Let $A > O$ be a positive matrix. Then

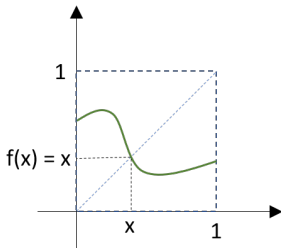
- A has a positive eigenvalue $\rho > 0$.
- The corresponding eigenvector X is positive as well and under condition $\sum x_i = 1$ it is unique (and is called **Perron vector**).
- For any other eigenvalue λ (real or complex) one have $|\lambda| < \rho$.
- There are no non-negative eigenvectors other than positive multiplies of the Perron vector.



Non-Negative Matrices

Brouwer's fixed point theorem: Every continuous map $f : K \mapsto K$ has a fixed point.

Ex. $f : [0, 1] \rightarrow [0, 1]$



Non-Negative Matrices

Brower's fixed point theorem: Every continuous map $f : K \mapsto K$ has a fixed point.

If $A \geq 0$ let $S = \{X \geq 0; \|X\|_2 = 1\}$ and $f : S \rightarrow S$,

$$f(X) = \frac{AX}{\|AX\|_2}.$$

Either there is $X \in S$ such that $AX = 0$ or f is cts. on $S \Rightarrow f$ has a fix point $X \Rightarrow \frac{AX}{\|AX\|_2} = X \Rightarrow AX = \|AX\|_2 X$

Conclusion: A has a non-negative eigenvector.



Primitive Matrices

Def. $A \geq 0$ is **primitive** if $A^k > 0$ for some k .

Ex. $P \geq 0$ is stochastic and primitive if all states can be reached in k steps.

Theorem 10.16

Let P be a primitive stochastic matrix. Then

- There exists a unique positive vector X such that $PX = X$ and $\sum x_i = 1$ and this is the Perron vector.
- There exists $\lim_{n \rightarrow \infty} P^n$.
- For each non-negative vector Y with $\sum y_i = 1$ one has $\lim_{n \rightarrow \infty} P^n Y = X$.



Graphs and Adjacency Matrices

Oriented graphs: Graph with n vertices (nodes), let a_{ij} = number of edges from v_i to v_j . $A = (a_{ij})$ is the adjacency matrix. $A \geq 0$.

Theorem 10.17

If $B = A^m$, then b_{ij} is equal to the number of different paths of length m in G from the vertex v_i to the vertex v_j . In particular $\text{tr } B$ is equal to the number of cyclic paths of length m .



PageRank

Each vertex v_i = a webpage

Edge = link from between v_i and v_j

$$a_{ij} = \begin{cases} 1 & \text{link} \\ 0 & \text{no link} \end{cases}, \quad a_{ii} = 1.$$

Approach 1: $R_i = \sum_j a_{ij} R_j \Rightarrow R = AR$

- R_i depends on the **number of links** and the ranks R_j of linking pages.
- A might not have eigenvalue 1.



PageRank

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Edge = link from between v_i and v_j

$$a_{ij} = \begin{cases} 1 & \text{link} \\ 0 & \text{no link} \end{cases}, \quad a_{ii} = 1.$$

Approach 2: $P = \left(\frac{a_{ij}}{\sum_k a_{kj}} \right)$, $R_i = \sum_j p_{ij} R_j \Rightarrow R = PR$

- R_j distributed over all links.
- P stochastic but might not be primitive.



PageRank

Each vertex v_i = a webpage

Edge = link from between v_i and v_j

$$a_{ij} = \begin{cases} 1 & \text{link} \\ 0 & \text{no link} \end{cases}, \quad a_{ii} = 1.$$

Approach 3: Add small number to each a_{ij} .

$$P = \left(\frac{a_{ij} + \epsilon}{\sum_k (a_{kj} + \epsilon)} \right), \quad R_i = \sum_j p_{ij} R_j \Rightarrow R = PR$$

- R_j distributed over all links.
- $P > 0$, stochastic, primitive.

Compute R using $\lim_{n \rightarrow \infty} P^n R_0$, R_0 = starting rank.



PageRank

Approach 3:

$$P = \begin{pmatrix} 1/3 & 1/9 & 1/9 \\ 1/3 & 4/9 & 4/9 \\ 1/3 & 4/9 & 4/9 \end{pmatrix}$$

With $Y = \frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ we get $PY = \begin{pmatrix} 0.1852 \\ 0.4074 \\ 0.4074 \end{pmatrix}$, $P^2Y = \begin{pmatrix} 0.1523 \\ 0.4239 \\ 0.4239 \end{pmatrix}$

$$X \approx \begin{pmatrix} 0.1429 \\ 0.4286 \\ 0.4286 \end{pmatrix}$$

